

Deep Generative Models

Chapter 4: Generative Adversarial Networks

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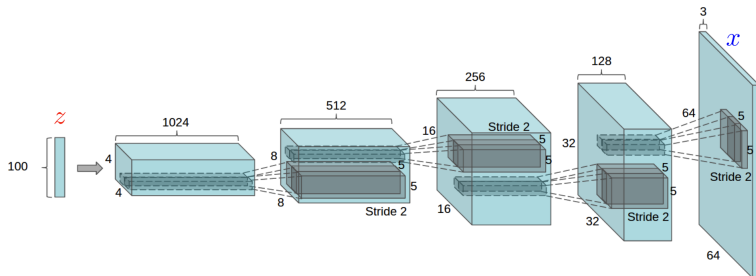
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Deep Convolutional GAN: Generator

DCGAN was one of basic architectures with a deep CNN generator¹



- A CNN discriminator is used with Sigmoid output

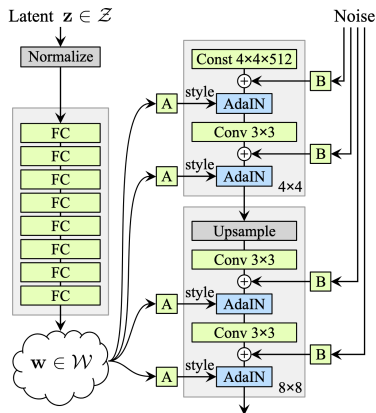
¹Proposed in 2016

DCGAN: *Sample Outputs*



Style GAN: Generator

Style GAN uses a more advanced generator²



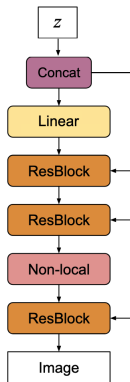
²Proposed in 2018

Style GAN: *Sample Outputs*



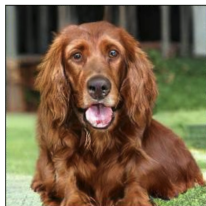
BigGAN: Generator

BigGAN uses residual blocks to build deeper generator³



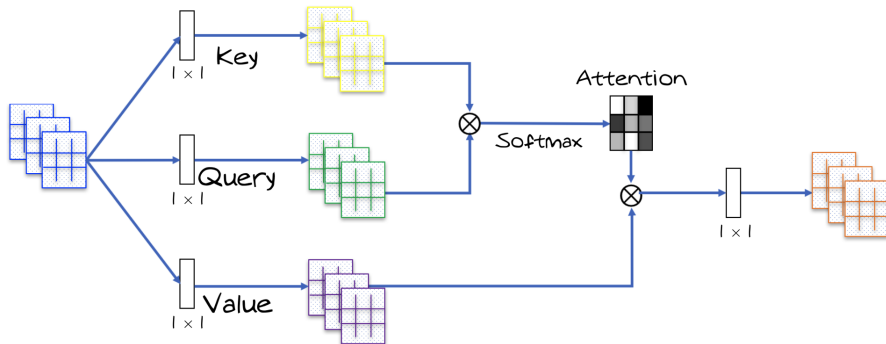
³Proposed in 2019

BigGAN: *Sample Outputs*



SAGAN: Generator

Self-Attention GAN uses attention mechanism at generator⁴



⁴Proposed in 2019

SAGAN: *Sample Outputs*



Wrap Up

GANs use **adversarial networks** for generation

- *Sampling is similar to flow-based models*
 - ↳ It is very **fast** and simple
 - ↳ **Latent** is typically **Gaussian**
- *For training, we solve a **min-max game***
 - ↳ **Vanilla** min-max $\equiv$ implicit **MLE**
 - ↳ **Wasserstein** min-max $\equiv$ implicit Wasserstein **distance minimization**

GANs have been a **real breakthrough**

- They are still being **state-of-the-art**
- They are pretty **straightforward** to implement

Next Stop: **Indirect** distribution learning by **variational inference**